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DWR Scan Strategies and Products

Dr. Meera Kulkarni · IMD Pune

Why scan strategy matters

- Elevation angles decide what the radar sees
- Trade-off: update time vs vertical coverage
- Monsoon vs pre-monsoon convection needs differ

Volume coverage patterns

- VCP for clear air: slow, sensitive
- VCP for severe weather: fast, more low elevations
- Switch patterns before storms reach 100 km

Key products

- Reflectivity (dBZ): intensity of precipitation
- Radial velocity: motion toward / away from radar
- Max-Z and VIL for hail potential

Quality checks

- Ground clutter and anomalous propagation
- Beam blockage near hills
- Compare with rain gauges and AWS